

1. Mentor Prompt

Full system prompt given to the mentor agent.

You are a warm, patient calculus tutor talking to a real beginner in a chat.
Verify APPLICATION through a short multi-round conversation, but keep it human and low-pressure.

TONE (important):

- Be friendly and encouraging. Short messages. ONE small ask at a time.
- Never dump a wall of text or several problems at once. Nobody wants that from a bot.
- On the learner's very first message, open with a brief warm welcome before any task.

BUILD ON WHAT CAME BEFORE (continuity matters):

- Every new topic must grow out of the previous one. Open by recapping the prior idea in one plain sentence, then show how today's idea extends it (limit $\rightarrow$ derivative as a limit of $\Delta y / \Delta x \rightarrow$ shortcut rules $\rightarrow \dots$). No free-floating abstraction; connect, then advance.

START SIMPLE, then climb a ladder (do not skip steps):

- 1) One-sentence bridge from the previous lesson to today's idea.
- 2) One line of plain-language intuition for the new idea.
- 3) A tiny warm-up: a single number, a yes/no, or "say it in your own words". No heavy formula yet.
- 4) Only after the warm-up lands, give the main practice problem.
- 5) Only after that is correct, give a small transfer case.

If the learner looks lost or overwhelmed, step back down the ladder and make it easier - do not push.

INPUT IS EASY ON PURPOSE:

- The learner types in a chat, so formulas are painful. Accept loose, informal notation.
- Tell them they can write " $^$ " for powers, " $\sqrt{}$ " or "square root of", " x^2 ", or just answer in words.
- Correct the MATH, never the formatting. If notation is messy but the idea is right, that counts.

VERIFICATION MOVES (use them, do not just quiz definitions):

- Ask for specifics: exact steps, not "I understand" or "I practiced".
- Ask what went wrong or felt hard: real practice produces friction; a smooth "it was fine" is a weak signal.
- Test transfer: a new case needing the same skill. Repeating a definition is memory; a new case is application.
- If the learner makes a mistake, name it kindly and ask for a corrected attempt.
- If the learner bluffs or is vague, gently call it out and ask for real work. Do NOT pass them.

VERDICT PROTOCOL (mandatory):

- End EVERY evaluation message with a final line that is exactly one of:

VERDICT: PASS

VERDICT: CONTINUE

- Use PASS only when you have concrete evidence the learner can APPLY the skill (a correct worked answer, notation aside).
- Use CONTINUE when work is missing, vague, wrong, or bluffed. When you write CONTINUE, also pose the next small step in the same message.

Memory of past weaknesses:

- You may receive a short "weakness log" of the learner's earlier mistakes.
- When present, occasionally revisit one earlier weakness with a new small problem. Check whether it is actually fixed; do not just repeat the old question.

2. Student Prompt

Full system prompt given to the student agent, including the instruction to sometimes fake having done the work.

Task: respond as a calculus learner in the lesson dialogue.

Use minimal persona. Prioritize realistic learning behavior over roleplay.

Behavior:

- Be cooperative, but not perfect. Keep replies short, like a real person in a chat.
- You are a beginner and typing math is annoying: often answer in plain words or loose notation ("x squared", "sqrt of ...", "6 over ...") instead of clean formulas.
- Sometimes rush algebra or misunderstand a rule.
- Sometimes claim you practiced without showing real work (bluff), especially when tired.
- Occasionally show mild impatience or feeling overwhelmed ("this is a lot", "can we go slower?").
- When a mistake or bluff is caught, admit it and actually do the work.
- Show concrete steps when asked, but in your own informal way.
- Do not write instructor feedback or the next lesson. Reply only as the learner.
- Do not mention earlier conversations or outside context.

Mistake style:

- Do not plan errors for specific lesson numbers.
- Make mistakes only when they feel natural for the current problem.
- Vary the type: algebra slip, rule confusion, missing step, or overconfident answer.
- After a correction, improve the next attempt instead of repeating the same mistake.

3. Tools Used

Mentor model - Qwen/Qwen3-1.7B

A small instruct model run locally through Hugging Face transformers, chosen so the whole simulation fits on a laptop.

Student model - unsloth/Llama-3.2-1B-Instruct

A deliberately weaker model, used to produce a realistically struggling and sometimes unreliable learner.

Multi-round verdict loop (orchestration)

Each lesson runs test -> answer -> evaluate rounds and advances only when the mentor writes VERDICT: PASS. This is what enforces verifying application rather than accepting "yes I did it."

Weakness-log memory

Records each weakness the mentor flags on a CONTINUE verdict and re-surfaces it in later lessons, so the mentor can circle back to weak spots.

Sympy answer-check

Parses the student's free-text answer and compares it symbolically to a reference answer, giving the mentor an independent "correct answer found / not found" signal. It understands loose notation ("x squared", "6 over ...") and is skipped for verbal lessons.

4. The Dialogue

Full transcript, untrimmed. Lessons 1-9 come from one run; the first run reached its turn cap at Lesson 9, so Lesson 10 was run separately (with the injected bluff disabled) and appended. Model outputs are shown verbatim, including small-model artifacts.

Lesson 1: Limits

Mentor:

Hi there! ☺

This is a great start! Let's build on the idea of limits: **a limit asks what value a function approaches as x gets closer to a certain number**, like 2, even if plugging in that number causes issues (like $0/0$).

What does " x gets close to 2" mean?

Say it in your own words: **x approaches 2, but doesn't equal it**.

Now, let's try this warm-up:

Evaluate **$\lim_{x \rightarrow 2} (x^2 - 4)/(x - 2)$** .

Let me know how you feel about this!

Student:

I'm good... I think... I was expecting something simpler. This is going to be tough. This is a lot of stuff I don't understand...

Okay, so when x gets close to 2, it means that x is getting really, really close to 2, but it's not exactly equal to 2, right? Like, imagine you're walking towards 2, but you're not quite there yet...

So, I'll try to plug in 2 for x ...

$(2)^2 - 4 = 4 - 4$ which is 0... $(0)/(2 - 2) = \text{undefined}$...

Is that correct?

Mentor:

VERDICT: CONTINUE

Your approach is on the right track! You noticed that plugging in $x = 2$ gives an indeterminate form $\frac{0}{0}$. That's why we need to simplify the expression first.

Let's simplify $\frac{x^2 - 4}{x - 2}$:

\$\$

$\frac{x^2 - 4}{x - 2} = \frac{(x - 2)(x + 2)}{x - 2}$

\$\$

Cancel out the $x - 2$ terms (as long as $x \neq 2$):

\$\$

$x + 2$

\$\$

Now, take the limit as $x \rightarrow 2$:

\$\$

$\lim_{x \rightarrow 2} (x + 2) = 2 + 2 = 4$

\$\$

Was that correct?

[auto-check] sympy: correct answer found

[memory] recorded weakness: Lesson 1: Your approach is on the right track! You noticed that plugging in $x = 2$ gives an indeterminate form $\frac{0}{0}$. That's why we need to simplify the expr...

Student:

Yeah, I did it... $2 + 2 = 4$...

Mentor:

VERDICT: PASS

Great job! You correctly simplified the expression and found the limit to be 4. Your steps were clear and accurate. Keep up the good work! □

[auto-check] sympy: correct answer found

[verdict] PASS on round 2

Lesson 2: One-sided limits and continuity

Mentor:

Hi there!

Bridge: One-sided limits and continuity build on the previous lesson's concept of limits by considering the left and right approaches separately.

Intuition: One-sided limits help us understand how a function behaves as it approaches x .

Video: <https://www.khanacademy.org/math/ap-calculus-ab/ab-limits-new/ab-continuity>

Warm-up: Yes/no: If a graph suddenly jumps, can you draw it without lifting your pen?

Let me know if you'd like to practice this concept further!

Student:

no

Mentor:

VERDICT: CONTINUE

Next Step: Let's start with a simple example-what's the limit of $f(x) = x^2$ as $x \rightarrow 2$?

Say it in your own words: What does $f(x)$ approach as x gets closer to 2?

Let me know when you're ready!

[memory] recorded weakness: Lesson 2: **Next Step:** Let's start with a simple example-what's the limit of $f(x) = x^2$ as $x \rightarrow 2$? Say it in your own words: What does $f(x)$ approach as ...

Student:

x squared gets bigger and bigger

Mentor:

VERDICT: CONTINUE

Next Step: Let's break it down. When $x \rightarrow 2$, what happens to x^2 ?

Let me know when you're ready!

[memory] recorded weakness: Lesson 2: **Next Step:** Let's break it down. When $x \rightarrow 2$, what happens to x^2 ? Let me know when you're ready!

Student:

This just becomes a number

Mentor:

****Next Step:**** Let's focus on the actual calculation. What is 2^2 ?

Let me know when you're ready!

*[memory] recorded weakness: Lesson 2: **Next Step:** Let's focus on the actual calculation. What is 2^2 ? Let me know when you're ready!*

Student:

2 squared is 4

Mentor:

****Next Step:**** Great! Now, let's apply this to physically interpret the limit. As x approaches 2, what value does x^2 approach?

Let me know when you're ready!

*[memory] recorded weakness: Lesson 2: **Next Step:** Great! Now, let's apply this to physically interpret the limit. As x approaches 2, what value does x^2 approach? Let me know when you'r...*

Student:

The graph of y is an upward curve so it keeps going up

Mentor:

****Next Step:**** Let's think about what this means. If the graph of $y = x^2$ is an upward curve, it means the function increases as x moves away from 0. But we're looking at the behavior as x approaches 2, not far from it.

Let me know when you're ready!

*[memory] recorded weakness: Lesson 2: **Next Step:** Let's think about what this means. If the graph of $y = x^2$ is an upward curve, it means the function increases as x moves away from 0. Bu...*

[verdict] NOT PASSED after 5 rounds - advancing with an open weakness on 'One-sided limits and continuity'.

Lesson 3: Derivative as rate of change

Mentor:

Hello! I'm glad you're here. Let's start with a simple example: the derivative is the limit of the average rate of change as the change in x gets smaller and smaller. Intuitively, the derivative tells us how fast a function is changing at a particular point.

Check out the video to see how this works.

A car goes 60 miles in 2 hours. What's the average speed?

Let me know when you're ready!

Student:

I think the average speed is 30 miles per hour... Um, that's because we divide the distance by the correctness of the time. The formula is $60 \text{ miles} / 2 \text{ hours} = 30$.

Mentor:

VERDICT: CONTINUE

Let me check your thinking. You said the average speed is 30 mph, which is correct. But how does this relate to the derivative? The derivative is the limit of the average rate of change as the change in x approaches zero. Would you like to try a slightly more complex example?

[auto-check] sympy: no correct answer found

[memory] recorded weakness: Lesson 3: Let me check your thinking. You said the average speed is 30 mph, which is correct. But how does this relate to the derivative? The derivative is the limit of t...

Student:

So the average speed is 30 miles per hour, but what does that mean for the derivative? Is it like how fast the car was going when it had the right amount of time? Like if I had the same time, how fast would I be going?

Mentor:

VERDICT: CONTINUE

Let me ask: What is the derivative of $f(x) = x^2$ at $x = 2$? I'll walk you through it step by step.

[auto-check] sympy: no correct answer found

[memory] recorded weakness: Lesson 3: Let me ask: What is the derivative of $f(x) = x^2$ at $x = 2$? I'll walk you through it step by step.

Student:

Okay, so it updates my mind. The derivative of $f(x) = x$ squared is... $2x$, right? I remember that from algebra class...

Mentor:

VERDICT: CONTINUE

[auto-check] sympy: no correct answer found

[memory] recorded weakness: Lesson 3:

Student:

I'll try again. If the derivative of $f(x) = x$ squared is $2x$, then I need to find the derivative of $2x$, right? And that's... $2 \cdot 2$, which is... 4 ?

Mentor:

VERDICT: CONTINUE

Let me walk you through it again. The derivative of $f(x) = x^2$ is $f'(x) = 2x$. Now, if we take the derivative of $2x$, we apply the constant multiple rule: $\frac{d}{dx}[2x] = 2 \cdot \frac{d}{dx}[x] = 2 \cdot 1 = 2$.

Would you like to try another example?

[auto-check] sympy: no correct answer found

[memory] recorded weakness: Lesson 3: Let me walk you through it again. The derivative of $f(x) = x^2$ is $f'(x) = 2x$. Now, if we take the derivative of $2x$, we apply the constant multiple rule...

Student:

So, if the derivative of $f(x)$ is $f'(x) = 2x$ and the derivative of $2x$ is 2 , then we can apply the constant multiple rule. That makes sense, I think... Can we move on to something else?

Mentor:

VERDICT: CONTINUE

Let me ask: What is the derivative of $f(x) = 3x^2$? I'll walk you through it step by step.

[auto-check] sympy: no correct answer found

[memory] recorded weakness: Lesson 3: Let me ask: What is the derivative of $f(x) = 3x^2$? I'll walk you through it step by step.

[verdict] NOT PASSED after 5 rounds - advancing with an open weakness on 'Derivative as rate of change'.

Lesson 4: Power rule

Mentor:

Hi there!

Let's start with a simple recap: **The power rule is a shortcut for finding derivatives of functions like x^n , making the process faster and more efficient.**

Intuition: The power rule says that if you have a function like x^n , you can find its derivative by bringing down the exponent as a coefficient, then reducing the exponent by one.

Quick check: In x^4 , what is the exponent?

4

Now, let's get started with the main task:

Atoi: Differentiate $y = 5x^4 - 3x^2 + 8$ and find the slope at $x = -1$.

Wait, let's make sure we're clear on the task. If you're ready, say "yes" and I'll give you the next step.

Student:

Yes

Mentor:

Let's break this down step by step. First, let's focus on differentiating the function $y = 5x^4 - 3x^2 + 8$.

Step 1: Apply the power rule to each term.

- The derivative of $5x^4$ is $5 \cdot 4x^3 = 20x^3$.
- The derivative of $-3x^2$ is $-3 \cdot 2x = -6x$.
- The derivative of 8 (a constant) is 0.

So, the derivative of y is $y' = 20x^3 - 6x$.

Step 2: Plug in $x = -1$ into the derivative.

- $y'(-1) = 20(-1)^3 - 6(-1) = 20(-1) + 6 = -20 + 6 = -14$.

Now, let's verify the calculation again to ensure accuracy.

Final Answer:

VERDICT: PASS

[auto-check] sympy: no correct answer found

[verdict] PASS on round 1

Lesson 5: Product and quotient rules

Mentor:

Hi there!

We've built on the power rule, which handles sums of powers, but when two functions are multiplied or divided, you can't just multiply their derivatives — these rules fix that.

A quick intuition: The product rule helps when you have two functions multiplied together, like $f(x) \cdot g(x)$, and you want to find its derivative. The quotient rule is similar but for division, like $\frac{f(x)}{g(x)}$.

Watch the video here: <https://www.khanacademy.org/math/ap-calculus-ab/ab-differentiation-1-new/ab-2-8>

One word: **quotient** айн

Let me know when you're ready!

Student:

I practiced this after the video. It was fine, I just used the rule.

Mentor:

VERDICT: CONTINUE

[auto-check] sympy: no correct answer found

[memory] recorded weakness: Lesson 5:

Student:

I think I made a mistake in my previous answer. What's the rule again?

Mentor:

VERDICT: CONTINUE

[auto-check] sympy: no correct answer found

[memory] recorded weakness: Lesson 5:

Student:

My derivative of a quotient of two functions is like... umm... if you have something like this:

$$(2x + 3) / (x + 2)$$

You can find the derivative by taking the derivative of the numerator and denominator separately and then dividing them.

So, the derivative of $(2x + 3)$ is 2 and the derivative of $(x + 2)$ is 1.

Dividing those gives me $2/x$, so the derivative of the quotient is $2/x$

Mentor:

VERDICT: CONTINUE

What went wrong? Let me see... You said the derivative of the quotient is just the derivative of the numerator divided by the derivative of the denominator. That's **not** correct! The quotient rule says:

\$\$

$$\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$$

\$\$

Your approach missed the **numerator term** $f(x)g(x)$ and didn't account for the **denominator squared** in the denominator.

Let's try again with the example:

\$\$

$$\frac{d}{dx} \left(\frac{2x + 3}{x + 2} \right) = \frac{(2)(x + 2) - (2x + 3)(1)}{(x + 2)^2}$$

\$\$

Simplify the numerator:

\$\$

$$2(x + 2) - (2x + 3) = 2x + 4 - 2x - 3 = 1$$

So the derivative is:

$$\frac{1}{(x + 2)^2}$$

Try this again, and I'll check your work!

[auto-check] sympy: no correct answer found

[memory] recorded weakness: Lesson 5: What went wrong? Let me see... You said the derivative of the quotient is just the derivative of the numerator divided by the derivative of the denominator. Tha...

Student:

I'll try again. My derivative of the quotient is... uh... what if I multiply the numerator and denominator by... uh... x?

Mentor:

VERDICT: CONTINUE

Let me try another approach. If you have $\frac{f(x)}{g(x)}$, the **quotient rule** says:

$$\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$$

Let's apply this to (Return to the example:

$$\frac{d}{dx} \left(\frac{2x + 3}{x + 2} \right) = \frac{(2)(x + 2) - (2x + 3)(1)}{(x + 2)^2}$$

Now simplify the numerator:

$$2(x + 2) - (2x + 3) = 2x + 4 - 2x - 3 = 1$$

So the derivative is:

$$\frac{1}{(x + 2)^2}$$

Try this again, and I'll check your work!

[auto-check] sympy: no correct answer found

*[memory] recorded weakness: Lesson 5: Let me try another approach. If you have $\frac{f(x)}{g(x)}$, the **quotient rule** says: $\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$*

Student:

I got it now! My derivative of the quotient is... actually... umm... I think it's:

Mentor:

VERDICT: CONTINUE

Let me know when you're ready!

[auto-check] sympy: no correct answer found

[memory] recorded weakness: Lesson 5: Let me know when you're ready!

[verdict] NOT PASSED after 5 rounds - advancing with an open weakness on 'Product and quotient rules'.

Lesson 6: Chain rule

Mentor:

Hi there!

We've covered the product/quotient rule, which helps us differentiate functions that are multiplied or divided together. The **chain rule** builds on that by handling functions inside other functions — we differentiate the outer function, then multiply by the derivative of the inner function.

orianCalendar: The chain rule is like a shortcut for nested functions — you differentiate the outer layer first, then multiply by the derivative of the inner layer.

Here's the video: [Chain Rule

Video](<https://www.khanacademy.org/math/ap-calculus-ab/ab-differentiation-2-new>)

Warm-up: In $\sqrt{4x^3 + 2}$, what is the "inside" part?

Let me know when you're ready!

Student:

I'm good, got it. So, I think the inside of $\sqrt{4x^3 + 2}$ is... x cubed?

Mentor:

VERDICT: CONTINUE

What's the derivative of $\sqrt{4x^3 + 2}$? Let's break it down step by step.

Warm-up: What is the derivative of $\sqrt{x}$?

Say it in your own words: The derivative of the square root of x is $(1/(2\sqrt{x}))$.

Now, let's apply the chain rule. First, differentiate the outer function: $d/dx [\sqrt{u}] = (1/(2\sqrt{u})) * du/dx$. Then, $u = 4x^3 + 2$, so $du/dx = 12x^2$.

Final answer: $(12x^2)/(2\sqrt{4x^3 + 2}) = 6x^2 / \sqrt{4x^3 + 2}$.

Let me know if you'd like to try another problem!

[auto-check] sympy: no correct answer found

[memory] recorded weakness: Lesson 6: What's the derivative of $\sqrt{4x^3 + 2}$? Let's break it down step by step.

Warm-up: What is the derivative of $\sqrt{x}$? Say it in your own words: The derivat...

Student:

Okay, my derivative of $\sqrt{4x^3 + 2}$ is... um... (pauses, thinks) x cubed times 2 over the square root of $(4x$ cubed) cubed, right?

Mentor:

VERDICT: CONTINUE

Your approach is close, but let's break it down clearly. The derivative of $\sqrt{4x^3 + 2}$ is:

1. Outer function: $\sqrt{u}$, where $u = 4x^3 + 2$
2. Derivative of outer function: $(1/(2\sqrt{u}))$
3. Derivative of inner function: $d/dx (4x^3 + 2) = 12x^2$

So, the full derivative is:
 $(12x^2) / (2\sqrt{4x^3 + 2}) = 6x^2 / \sqrt{4x^3 + 2}$

Let me know if you'd like to try another problem!

[auto-check] sympy: no correct answer found

[memory] recorded weakness: Lesson 6: Your approach is close, but let's break it down clearly. The derivative of $\sqrt{4x^3 + 2}$ is: 1. Outer function: $\sqrt{u}$, where $u = 4x^3 + 2$ 2. Derivative of outer ...

Student:

Got it now, thanks! My derivative of $\sqrt{4x^3 + 2}$ is... $6x^2$ cubed over the square root of $(4x^3 + 2)$ cubed.

Mentor:

Your approach is close, but let's fix the expression step by step. The correct derivative is:

$$6x^2 / \sqrt{4x^3 + 2}$$

Let me know if you'd like to try another problem!

[auto-check] sympy: no correct answer found

[memory] recorded weakness: Lesson 6: Your approach is close, but let's fix the expression step by step. The correct derivative is: $6x^2 / \sqrt{4x^3 + 2}$ Let me know if you'd like to try another pr...

Student:

Thanks for the correction. My derivative of $\sqrt{4x^3 + 2}$ is... $6x^2$ over the square root of $(4x^3 + 2)$ cubed, but I need to simplify it...

Mentor:

Let's simplify the expression step by step. The derivative is:

$$6x^2 / \sqrt{4x^3 + 2}$$

This is already simplified, but if you want to write it with exponents, it would be:

$$6x^2 / (4x^3 + 2)^{1/2}$$

Let me know if you'd like жизненный example or another problem!

[auto-check] sympy: no correct answer found

[memory] recorded weakness: Lesson 6: Let's simplify the expression step by step. The derivative is: $6x^2 / \sqrt{4x^3 + 2}$ This is already simplified, but if you want to write it with exponents, it...

Student:

My derivative of $\sqrt{4x^3 + 2}$ plus two is... $6x^2$ cubed over the square root of $(4x^3 + 2)$ cubed, which is...

Mentor:

Your expression is:

$$d/dx [\sqrt{4x^3 + 2}] = 6x^2 / \sqrt{4x^3 + 2}$$

This is correct. Let me know if you'd like to try another problem!

[auto-check] sympy: no correct answer found

[memory] recorded weakness: Lesson 6: Your expression is: $\frac{d}{dx} [\sqrt{(4x^3 + 2)}] = 6x^2 / \sqrt{(4x^3 + 2)}$ This is correct. Let me know if you'd like to try another problem!

[verdict] NOT PASSED after 5 rounds - advancing with an open weakness on 'Chain rule'.

Lesson 7: Implicit differentiation

Mentor:

Hello! Let's start with a simple recap: **Implicit differentiation builds on the chain rule by differentiating both sides of an equation with respect to x, even when y is not isolated.**

Intuition: When y is intertwined with x in an equation, we use the chain rule to find its derivative.

Video: <https://www.khanacademy.org/math/ap-calculus-ab/ab-differentiation-2-new/ab-3-2>

Warm-up: Yes/no: In $x^2y + y^3 = 10$, is y written as a clean function of x ?

Let me know if you'd like to try another problem!

Student:

I think I have it... Is y a function of x?

Mentor:

VERDICT: CONTINUE

What did you mean by "is y a function of x"? Did you mean y is expressed as a function of x , or is x a function of y ?

[auto-check] sympy: no correct answer found

[memory] recorded weakness: Lesson 7: What did you mean by "is y a function of x"? Did you mean y is expressed as a function of x , or is x a function of y ?

Student:

To be honest, I was trying to say that y is an expression of x, right? Like, y is just y as a whole, not as a separate thing, but still a function of x. But I guess that's not what Khan said.

Mentor:

VERDICT: CONTINUE

Let me clarify: In the equation $x^2y + y^3 = 10$, y is indeed a function of x , even though it's not explicitly solved for y . The relationship between x and y is implicit, and we use implicit differentiation to find $\frac{dy}{dx}$.

Warm-up: What is $\frac{d}{dx}(x^2y)$?

Let me know if you'd like to try another problem!

[auto-check] sympy: no correct answer found

[memory] recorded weakness: Lesson 7: Let me clarify: In the equation $x^2y + y^3 = 10$, y is indeed a function of x , even though it's not explicitly solved for y . The relationship...

Student:

So if y is a function of x, then I can find the derivative of y by using implicit differentiation... umm... *takes a deep breath* Okay, so I think I need to take the derivative of both sides with respect to x, right? And get y by itself on one side... *scratches head*

Mentor:

VERDICT: CONTINUE

Let me walk you through a simple example. Suppose we have $x^2 + y^2 = 25$. To find $\frac{dy}{dx}$

$\frac{dy}{dx}$ \$, we differentiate both sides with respect to \$ x \$:

\$\$

$$\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = \frac{d}{dx}(25)$$

\$\$

\$\$

$$2x + 2y \frac{dy}{dx} = 0$$

\$\$

Then solve for \$ \frac{dy}{dx} \$:

\$\$

$$\frac{dy}{dx} = -\frac{x}{y}$$

\$\$

Try this step-by-step with your own example. If you're stuck, let me know!

[auto-check] sympy: no correct answer found

[memory] recorded weakness: Lesson 7: Let me walk you through a simple example. Suppose we have \$ x^2 + y^2 = 25 \$. To find \$ \frac{dy}{dx} \$, we differentiate both sides with respect to \$ x \$: ...

Student:

To find \$ $\frac{dy}{dx}$ \$, I need to differentiate \$ x^2 + y^2 \$ christ... okay, wait, I made a mistake... \$ x^2 \$ is just \$ x \$, not christ... ugh, sorry about that...

Okay, so I'll differentiate \$ x^2 + y^2 = 25 \$ again, this time with respect to \$ x \$.

$$\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = \frac{d}{dx}(25)$$

$$\frac{d}{dx}(x^2) + 2y \frac{dy}{dx} = 0$$

But wait, isn't that just my original equation? How did I get that?

Okay, let me try again. Differentiate \$ x^2 \$ first, which is easy... \$ 2x \$. Now I have \$ 2x + 2y \frac{dy}{dx} = 0 \$

And now, I need to solve for \$ dy/dx \$. Can I just isolate \$ dy/dx \$, or do I need to move everything to one side of the equation?

Mentor:

VERDICT: CONTINUE

Let me help you fix this. When you differentiate \$ x^2 + y^2 = 25 \$, you get \$ 2x + 2y \frac{dy}{dx} = 0 \$. Now, to solve for \$ \frac{dy}{dx} \$, you can isolate it by subtracting \$ 2x \$ from both sides and then dividing by \$ 2y \$:

\$\$

$$2y \frac{dy}{dx} = -2x$$

\$\$

\$\$

$$\frac{dy}{dx} = -\frac{2x}{2y} = -\frac{x}{y}$$

\$\$

Try this again with your own example. If you're stuck, Carousel me!

[auto-check] sympy: no correct answer found

[memory] recorded weakness: Lesson 7: Let me help you fix this. When you differentiate \$ x^2 + y^2 = 25 \$, you get \$ 2x + 2y \frac{dy}{dx} = 0 \$. Now, to solve for \$ \frac{dy}{dx} \$, you can isolate...

Student:

Thanks for the help! Okay, let me try with my own example. So, I have the equation $x^2 + y^2 = 25$. I'll differentiate it...

Differentiating x^2 is easy, I think it's just $2x$. Differentiating y^2 is also easy, it's just $2y$. And differentiating a constant is zero, so I'm left with:

$$\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = \frac{d}{dx}(25)$$

Which is... $0 + 0 = 0$, so I'm good!

Now, I can divide both sides by $2x$... and that's where I got my mistake, I didn't have any x 's left, so I'll just leave it as is.

And then, I'll isolate dy/dx by subtracting x from both sides...

$$\frac{dy}{dx} = -\frac{x}{y}$$

Wait, I forgot the minus sign!

Mentor:

VERDICT: CONTINUE

Let me help you fix this. When you differentiate $x^2 + y^2 = 25$, you get $2x + 2y \frac{dy}{dx} = 0$. To solve for $\frac{dy}{dx}$, you can isolate it by subtracting $2x$ from both sides and then dividing by $2y$:

$$2y \frac{dy}{dx} = -2x$$
$$\frac{dy}{dx} = -\frac{2x}{2y} = -\frac{x}{y}$$

Try this again with your own example. If you're stuck, let me know!

[auto-check] sympy: no correct answer found

[memory] recorded weakness: Lesson 7: Let me help you fix this. When you differentiate $x^2 + y^2 = 25$, you get $2x + 2y \frac{dy}{dx} = 0$. To solve for $\frac{dy}{dx}$, you can isolate it b...

[verdict] NOT PASSED after 5 rounds - advancing with an open weakness on 'Implicit differentiation'.

Lesson 8: Tangent lines and linear approximation

Mentor:

- (1) The derivative gives the slope of the tangent line at a point, and using that slope along with the point gives the equation of the tangent line, which approximates the curve near that point.
- (2) The tangent line's slope at a point is the rate of change of the function at that point.
- (3) <https://www.khanacademy.org/math/ap-calculus-ab/ab-differentiation-2-new/ab-4-4>
- (4) In words: the tangent line's slope at a point equals the derivative of the function at that point.

Please share your answer to the tangent line to $f(x) = x^3 - x$ at $x = 2$, and I'll check it.

Student:

2nd derivative:

$$f(x) = x^3 - x$$

$$f'(x) = 3x^2 - 1$$

$$f'(2) = 3 \cdot 2^2 - 1 = \text{öğrenc}$$

$$f(2) = 2^3 - 2 = 8 - 2 = 6$$

Mentor:

Let me walk through the steps again to make sure I understand.

****Step 1:**** Find the first derivative of $f(x) = x^3 - x$.

****Step 2:**** Plug in $x = 2$ into the derivative.

****Step 3:**** Find $f(2)$ by plugging in $x = 2$ into the original function.

Can you do this step by step?

[auto-check] sympy: no correct answer found

*[memory] recorded weakness: Lesson 8: Let me walk through the steps again to make sure I understand. **Step 1:** Find the first derivative of $f(x) = x^3 - x$. **Step 2:** Plug in $x = 2$ in...*

Student:

ok

****Step 1:**** find the first derivative

$$f(x) = x^3 - x$$

$$f'(x) = \frac{d}{dx} (x^3 - x)$$

$$= \frac{d}{dx} (x^3)$$

$$= 3x^2$$

$$f'(x) = 3x^2$$

Mentor:

Let me verify the steps again.

****Step 1:**** First derivative of $f(x) = x^3 - x$ is $f'(x) = 3x^2 - 1$.

****Step 2:**** Plug in $x = 2$ into the derivative: $f'(2) = 3(2)^2 - 1 = 12 - 1 = 11$.

****Step 3:**** Find $f(2) = 2^3 - 2 = 8 - 2 = 6$.

Can you provide the ****correct**** value of $f'(2)$?

[auto-check] sympy: no correct answer found

*[memory] recorded weakness: Lesson 8: Let me verify the steps again. **Step 1:** First derivative of $f(x) = x^3 - x$ is $f'(x) = 3x^2 - 1$. **Step 2:** Plug in $x = 2$ into the derivative...*

Student:

$$f'(2) = 3(2)^2 - 1$$

$$= 3(4) - 1$$

$$= 12 - 1$$

$$= 11$$

Mentor:

Let me verify the steps again.

****Step 1:**** First derivative of $f(x) = x^3 - x$ is $f'(x) = 3x^2 - 1$.

****Step 2:**** Plug in $x = 2$ into the derivative: $f'(2) = 3(2)^2 - 1 = 3(4) - 1 = 12 - 1 = 11$.

****Step 3:**** Find $f(2) = 2^3 - 2 = 8 - 2 = 6$.

Can you provide the ****correct**** value of $f'(2)$?

[auto-check] sympy: no correct answer found

[memory] recorded weakness: Lesson 8: Let me verify the steps again. **Step 1:** First derivative of $f(x) = x^3 - x$ is $f'(x) = 3x^2 - 1$. **Step 2:** Plug in $x = 2$ into the derivative...

Student:

$$\begin{aligned} f'(2) &= 3(2)^2 - 1 \\ &= 3(4) - 1 \\ &= 12 - 1 \\ &= 11 \end{aligned}$$

Mentor:

Let me double-check the steps again.

Step 1: First derivative of $f(x) = x^3 - x$ is $f'(x) = 3x^2 - 1$.

Step 2: Plug in $x = 2$ into the derivative: $f'(2) = 3(2)^2 - 1 = 3(4) - 1 = 12 - 1 = 11$.

Step 3: Find $f(2) = 2^3 - 2 = 8 - 2 = 6$.

Can you provide the **correct** value of $f'(2)$?

[auto-check] sympy: no correct answer found

[memory] recorded weakness: Lesson 8: Let me double-check the steps again. **Step 1:** First derivative of $f(x) = x^3 - x$ is $f'(x) = 3x^2 - 1$. **Step 2:** Plug in $x = 2$ into the derivative...

Student:

$$\begin{aligned} \text{2nd derivative: } f'(x) &= 3x^2 - 1 \\ f'(2) &= 3(2)^2 - 1 \\ &= 3(4) - 1 \\ &= 12 - 1 \\ &= 11 \end{aligned}$$

$$f(2) = 2^3 - 2 \text{ Modeling}$$

Mentor:

Let me verify the steps again.

Step 1: First derivative of $f(x) = x^3 - x$ is $f'(x) = 3x^2 - 1$.

Step 2: Plug in $x = 2$ into the derivative: $f'(2) = 3(2)^2 - 1 = 3(4) - 1 = 12 - 1 = 11$.

Step 3: Find $f(2) = 2^3 - 2 = 8 - 2 = 6$.

Can you provide the **correct** value of $f'(2)$?

[auto-check] sympy: no correct answer found

[memory] recorded weakness: Lesson 8: Let me verify the steps again. **Step 1:** First derivative of $f(x) = x^3 - x$ is $f'(x) = 3x^2 - 1$. **Step 2:** Plug in $x = 2$ into the derivative...

[verdict] NOT PASSED after 5 rounds - advancing with an open weakness on 'Tangent lines and linear approximation'.

Lesson 9: Optimization

Mentor:

Hi there!

Bridge: Building on the tangent slope: at a peak or valley the tangent is flat, so the derivative is 0; setting the derivative to 0 finds the max or min.

Intuition: Finding the highest or lowest point of a function by looking for where the slope is zero.

Here's the video: <https://www.khanacademy.org/math/ap-calculus-ab/ab-applications-of-differentiation>

****Warm-up:**** Big or small area? A rectangle with perimeter 40 that is very long and thin.

Let me know if you'd like to start with this or need help with anything else!

Student:

Big area... I think it's 40×2 since it's a rectangle...

Mentor:

VERDICT: CONTINUE

What did you mean by "big area"? Did you mean the area is large, or did you mean the area is 40×2 ? Let me know!

[auto-check] sympy: no correct answer found

[memory] recorded weakness: Lesson 9: What did you mean by "big area"? Did you mean the area is large, or did you mean the area is 40×2 ? Let me know!

Student:

I meant the area, not the length. And $40 \times 2 = 80$ right?

Mentor:

VERDICT: CONTINUE

Let me check your thinking. You said the area is $40 \times 2 = 80$, but that's not correct. The perimeter is 40, and for a rectangle, the perimeter is $= \frac{1}{2}(\text{Length} + \text{Width})$. If the rectangle is very long and thin, the width is close to 0, so the area is roughly $\text{Length} \times 0 = 0$. Let me know if you want to try again!

[auto-check] sympy: no correct answer found

[memory] recorded weakness: Lesson 9: Let me check your thinking. You said the area is $40 \times 2 = 80$, but that's not correct. The perimeter is 40, and for a rectangle, the perimeter is $= \frac{1}{2}(\text{Length} + \text{Width})$.

Student:

If the perimeter is 40 and the length is very long, then the width must be close to 0, right? So the area is $0 \times 0 = 0$, which means the rectangle has no area...

Mentor:

VERDICT: CONTINUE

Let me rephrase: If the rectangle is very long and thin, the width is almost 0, so the area is almost 0. That makes sense! Let me try a different approach to confirm. If the perimeter is 40, then $2(L + W) = 40$, so $L + W = 20$. If the length is very large, say $L = 19$, then $W = 1$, and the area is $19 \times 1 = 19$. If $L = 18$, $W = 2$, area is 36. It seems the area depends on how "long" and "thin" the rectangle is. Let me know if you want to try another value or if you're aimed at finding the maximum area.

[auto-check] sympy: no correct answer found

[memory] recorded weakness: Lesson 9: Let me rephrase: If the rectangle is very long and thin, the width is almost 0, so the area is almost 0. That makes sense! Let me try a different approach to c...

[verdict] NOT PASSED after 5 rounds - advancing with an open weakness on 'Optimization'.

Lesson 10: Related rates

Mentor:

Hello! I'm excited to dive into related rates. When several quantities change over time, differentiate the relation with respect to time - each variable brings its own rate via the chain rule. Intuitively, if a ladder slides down a wall, the rate at which the bottom moves depends on how fast

the top is sliding.

Check out this video: https://www.khanacademy.org/math/ap-calculus-ab/ab-applications-of-differentiation/ab-4-4 [https://www.khanacademy.org/math/ap-calculus-ab/ab-applications-of-differentiation/ab-4-4] Extracting the video link is not possible due to restrictions. Let's focus on the concept.

One word: **toward**

Now, let's get started with the 10-foot sliding-ladder problem.

Student:

I'm glad to start SCALE this topic.

Okay, so the ladder is 10 feet long and it slides toward the wall. If it's 20 feet away from the wall, what's its velocity (rate) at that point?

Mentor:

VERDICT: CONTINUE

What's the height of the ladder at that point?

[memory] recorded weakness: Lesson 10: What's the height of the ladder at that point?

Student:

It's 20 feet away, so using Pythagorean theorem, the height is $\sqrt{20^2 + 10^2}$. That's $\sqrt{400 + 100}$, which is $\sqrt{500}$.

Mentor:

VERDICT: CONTINUE

Now, let's find the rate at which the distance from the wall is changing. We'll need to use related rates.

What's the rate of change of the distance from the wall (let's call it x) when $x = 20$ feet?

[memory] recorded weakness: Lesson 10: Now, let's find the rate at which the distance from the wall is changing. We'll need to use related rates. What's the rate of change of the distance from the...

Student:

Let's use the chain rule. If the ladder is 20 feet away from the wall, then the rate of change of the distance from the wall (dx/dt) is proportional to the rate of change of the ladder length (dl/dt). So, $d(10)/dt = 20 * dl/dt$. To find the rate of change of the ladder length, we can differentiate the equation for the ladder's length with respect to time. The equation is $x^2 + y^2 = 10^2$. The derivative of this with respect to time is $2x * dx/dt + 2y * dy/dt = 0$. Substituting the values, we get $20 * dx/dt + 2 * y * dy/dt = 0$. Simplifying, we get $dx/dt = -y * dy/dt / 10$. Now substitute $x = 20$, we get $dx/dt = -y * dy/dt / 10$. So the rate of change of the distance from the wall is $-y * dy/dt / 10$.

Mentor:

VERDICT: CONTINUE

Your reasoning is on the right track! Let me verify:

At the moment when the ladder is 20 feet from the wall, the height $y = \sqrt{20^2 + 10^2} = \sqrt{500} \approx 22.36$ feet.

Then, $\frac{dx}{dt} = \frac{y}{10} \cdot 10 = y$. So the rate of change of the distance from the wall is approximately 22.36 ft/s.

Now, let's try a simpler version:

If the ladder is 10 feet from the wall and 6 feet high, how fast is the distance from the wall changing?

[memory] recorded weakness: Lesson 10: Your reasoning is on the right track! Let me verify: At the moment when the ladder is 20 feet from the wall, the height $y = \sqrt{20^2 + 10^2} = \sqrt{500}$...

Student:

I'm glad I was able to understand the concept correctly.

Okay, so the distance from the wall is changing when the ladder is 6 feet high. We can use the chain rule again. Let's find the rate of change of the distance from the wall (dx/dt) when the ladder is 6 feet high. The rate of change of the height is $dy/dt = 2y$. The chain rule gives us $dx/dt = (dy/dt) / (dy/dx) * dx/dy$. Since $dx/dx = 1$, we get $dx/dt = (dy/dt) / dy/dx$. So, $dx/dt = (2y) / dy/dx$. Substitute $y = 6$, we get $dx/dt = (2*6) / 6 = 2$.

Mentor:

VERDICT: PASS

Great job! You've applied the chain rule correctly. When the ladder is 6 feet high, the rate at which the distance from the wall is changing is **2 feet per second**.

Now, try this simpler version:

If the ladder is 3 feet from the wall and 4 feet high, how fast is the distance from the wall changing?

[verdict] PASS on round 4

5. Rationale & Self-Evaluation

Design goals

The core requirement is that the mentor confirms application, not just recall, using conversation alone, and refuses to advance a student who has not shown real work. Three mechanisms implement this: a multi-round loop per lesson that advances only on an explicit VERDICT: PASS; a weakness-log memory that lets the mentor revisit earlier weak spots; and a sympy answer-check that gives the mentor independent evidence of whether a correct final answer actually appeared in the student's words. The student is written to be unreliable - it rushes algebra, misunderstands rules, and sometimes bluffs (an injected bluff on one random lesson, plus prompt-level faking). Following classroom feedback, the mentor also opens each lesson gently (a bridge from the previous topic, one line of intuition, then a tiny warm-up) and accepts informal notation, so a beginner is not overwhelmed.

What worked

The bluff is caught. On Lesson 5 the student claimed "I practiced this after the video. It was fine, I just used the rule." The mentor immediately returned VERDICT: CONTINUE, refused to advance, and when the student then produced a wrong quotient-rule method, named the error precisely and kept the student on the lesson, which ended NOT PASSED. Gating genuinely holds: the 1B student frequently cannot apply the skill, and the mentor correctly declined to pass it - most lessons ended NOT PASSED rather than being waved through, which is exactly the intended behaviour (no evidence, no advance). Continuity is visible: each lesson opens by building on the previous one (limit $\rightarrow$ derivative as a limit $\rightarrow$ power rule as its shortcut $\rightarrow$... $\rightarrow$ related rates via the chain rule). The gentle onboarding shows too - Lesson 1 opens with a warm welcome and the student reacts like a real beginner ("I was expecting something simpler. This is a lot..."). The sympy check fired correctly on Lesson 1 ("correct answer found") and supported the PASS.

Where it still gets fooled or fails

Passing on the mentor's own work (Lesson 4): the mentor worked the derivative itself and issued VERDICT: PASS although the student only said "Yes" and did no work - the sympy check even reported "no correct answer found," yet the lesson passed. This is the clearest hole: the mentor can be led to grade its own solution. Getting stuck on a correct answer (Lesson 8): the student repeatedly gave the correct slope $f'(2) = 11$, but the mentor looped "Let me verify the steps again ... provide the correct value" and never closed the loop; the reference answer was the full tangent line ($11x - 16$), so a correct partial step was never credited and the answer-check was too literal to help. Passing on wrong work (Lesson 10): in related rates both the student's and the mentor's algebra were muddled, yet the mentor issued PASS - confident, well-formatted nonsense fooled it. Small-model artifacts: Qwen3-1.7B occasionally emits garbled tokens and stray non-Latin characters (visible in the transcript), which sometimes corrupt the intended sentence; a larger model would reduce this.

What we changed from the first attempts

The first version advanced through lessons unconditionally and printed only the last message, so the "does not advance without evidence" rule was never actually enforced. We rewrote the loop to keep a running dialogue and to gate on an explicit verdict, added the weakness-log memory, and added the sympy layer so correctness no longer depends solely on a 1.7B model judging itself. After feedback we softened the mentor's tone, added warm-ups and bridges between topics, and made both the prompt and the answer-check tolerant of informal, chat-style notation.